

Literature Status: Random Regular Graphs as Nonlinear Expanders with Respect to Each Other

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Status: literature already answered. Question 2.4 of Mendel–Naor (arXiv:1306.5434) asks whether two independent random cubic graphs have a dimension-free nonlinear Poincaré constant. Altschuler–Dodos–Tikhomirov–Tyros (arXiv:2506.17433, Theorem 1.3) explicitly give a complete affirmative answer, with arbitrary fixed degrees and all exponents $p \geq 1$.

1 Original question

Let $\mathcal{G}_{n,d}$ denote the uniform probability distribution on simple d -regular graphs on n vertices. For a graph H , write d_H for its shortest-path metric, and let $\gamma(G, d_H^2)$ be the least constant in the corresponding nonlinear Poincaré inequality.

In Section 2.1, Question 2.4, on printed page 11 of *Expanders with respect to Hadamard spaces and random graphs*, Mendel and Naor ask [1]:

Source Question 2.4 (Kleinberg). *Does there exist a universal constant $K < \infty$ such that*

$$\lim_{\min\{m,n\} \rightarrow \infty} (\mathcal{G}_{m,3} \times \mathcal{G}_{n,3}) \{ (G, H) : \gamma(G, d_H^2) \leq K \} = 1?$$

Thus the question permits different vertex-set sizes and quantifies over all maps from the vertices of G into the graph metric of H . The source proves only the special case of bijections between equal-size graphs in Proposition 8.1.

2 Separate later answer

Altschuler, Dodos, Tikhomirov, and Tyros explicitly restate this as Problem 1.2 of *Discrete Poincaré inequalities and universal approximators for random graphs* [2]. Their abstract calls their result a complete affirmative resolution of Kleinberg’s problem. Theorem 1.3, on printed page 3, proves the following stronger statement.

Answering Theorem 1.3. *For fixed integers $d, \Delta \geq 3$ and every $p \geq 1$, there are constants $\Gamma(d, p) < \infty$ and $\tau(d, \Delta) > 0$ such that independent $G \sim \mathcal{G}_{n,d}$ and $H \sim \mathcal{G}_{m,\Delta}$ satisfy*

$$\gamma(G, d_H^p) \leq \Gamma(d, p)$$

simultaneously for every $p \geq 1$, with probability at least $1 - O_{d,\Delta}(\min\{n, m\}^{-\tau})$.

Taking $d = \Delta = 3$ and $p = 2$ gives the source statement with $K = \Gamma(3, 2)$, because the displayed error tends to zero as $\min\{m, n\} \rightarrow \infty$. The match is exact and author-explicit, not an agent-inferred reformulation.

3 Scope and search evidence

The later theorem completely answers Question 2.4 and also strengthens it to arbitrary fixed degrees and all $p \geq 1$. It does not settle the source paper’s distinct union-stability question for nonlinear spectral gaps in the remark on printed page 26.

A bounded search through 11 August 2026 checked the run’s registry and solution indexes, the exact wording “two stochastically independent random 3-regular graphs”, the source title and citations, and recent work on random-graph nonlinear spectral gaps. The decisive match is arXiv:2506.17433v2, whose Problem 1.2 cites Mendel–Naor Question 2.4 and whose Theorem 1.3 is labelled an affirmative solution. No new mathematical result is claimed here.

References

- [1] M. Mendel and A. Naor, *Expanders with respect to Hadamard spaces and random graphs*, Duke Math. J. **164** (2015), 1471–1548; arXiv:1306.5434.
- [2] D. J. Altschuler, P. Dodos, K. Tikhomirov, and K. Tyros, *Discrete Poincaré inequalities and universal approximators for random graphs*, arXiv:2506.17433v2 (2025).